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Optimal G^2 Hermite interpolation for 3D curves*R. Herzog^{1,*}, Ph. Blanc¹*University of Applied Sciences HEIG-VD, route de Cheseaux 1, CH-1401
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Abstract

We consider a Hermite interpolation problem for a 3D curve where the functional to be minimized is defined as the integral of squared norm of the third parametric derivative, subject to G^2 continuity constraints at the end points. The first order necessary optimality condition of the variational problem leads to a parametric transition curve with quintic polynomials. The determination of coefficients is given by a polynomial system with 2 unknowns. Stationary points correspond to positive roots of the resultant which is a degree 9 polynomial.

Although the formulated variational problem is non-convex, the proposed approach leads to the global solution, which can be computed in a reliable and fast manner.

Keywords: geometric G^2 continuity, Hermite 3D interpolation, curve completion, variational problem, quintic polynomial

1. Introduction

The following work was originally motivated by applications in CNC machining, but it can as well be used for CAD/CAM applications. Programmed paths in CNC machining consist of a large number of curve pieces in 3D space with non-smooth junction points. A common approach in CNC consists of breaking up the problem in two subsequent steps: geometry optimization [1, 2, 3] followed by time optimization (feedrate planning). The present paper addresses the first step, geometry optimization.

The geometrical discontinuities in the programmed path must be removed by rounding in order to prevent productivity decrease and a potential excitation of machine vibrations. Therefore, a part of the abutted curves on both sides of the discontinuity should be cut out, and the missing parts should be completed

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by a transition curve with geometrical G^2 continuity at the points where the transition curve rejoins the programmed path.

This problem is known as G^2 Hermite interpolation, and many different solutions have been proposed in the past. The wanted transition curve should be cusp-free, loop-free, and 'fair'. Existing solutions can be divided in 2 groups.

In the first group, an 'ad hoc' transition curve is imposed, using e.g. arcs of clothoids (Euler spirals) [4, 5, 6, 7]. The junction point of the clothoid arcs can be calculated for different cases of curve completion: line-line, line-circle and circle-circle. A 3D generalization of clothoids with G^1 continuity is presented in [8], where both curvature and torsion evolve linearly w.r.t. arc length.

In the second group, the transition curve is found by formulating a variational problem. Most existing rounding methods are optimization based problems using either B-spline or NURBS parametrizations. In [3] a method is presented for both rounding and compressing multiple small line segments. The resulting optimization problem is non-convex and is solved using a general purpose NLP solver. Minimizing strain energy under G^1 constraints at the end points was proposed in [9]. In order to assure the continuity of curvature, it is more natural to minimize the integral of squared change rate of curvature. This criterion was used in [10, 11, 12] using a rational cubic parametrization. This allows G^2 Hermite interpolation with circular precision. However the methods mentioned before are basically limited to planar curves.

The method presented in this paper is intended for 3D curves, where the variational problem is solved analytically without need for approximating the solution by discretization.

The paper is organized as follows. Section 2 recalls the notion of geometric G^2 continuity and shows its advantage over parametric C^2 continuity. Section 3 sets up the rounding problem formulation. The analytical solution is presented in Section 4, and the corresponding algorithm is explained in Section 5. Section 6 gives a numerical example related to the well known problem of G^2 circular Hermite interpolation. The choice of lift-off points, the only tuning parameter in the proposed rounding algorithm, is analyzed in Section 7. A comparison between the proposed polynomial rounding and clothoids is shown in Section 8. Section 9 concludes with a summary of features of the proposed rounding method.

2. Motivation for G^2 continuity

Parametric transition curves are often constructed such that the first n parametric derivatives agree with those of the curve segments abut. This C^n continuity disallows many parametrizations of smooth geometric transitions [13]. Geometric G^n continuity concerns only shape, independent from parametrization. Here we focus on G^2 continuity, defined for regular curves $\mathbf{r}'(u) \neq \mathbf{0}$ as follows:

Definition 1. According to [13], the junction point of two piecewise defined C^2 curves is called G^2 continuous if the 1st parametric derivatives at the junction

are collinear, and the curvature vector, defined as $\kappa \mathbf{n}$ according to Equation (3), is continuous.

In order to motivate G^2 continuity, we first consider the following C^2 interpolation example, linking $\mathbf{p}_0 = (1, 0)$ to $\mathbf{p}_1 = (0, 1)$, with 1st parametric derivatives $\mathbf{v}_0 = (0, 5)$, $\mathbf{v}_1 = (-5, 0)$, and 2nd parametric derivatives $\mathbf{a}_0 = \mathbf{a}_1 = \mathbf{0}$. A unique quintic polynomial $\mathbf{r}(u)$ fits the six interpolation conditions $\mathbf{r}(0) = \mathbf{p}_0$, $\mathbf{r}'(0) = \mathbf{v}_0$, $\mathbf{r}''(0) = \mathbf{a}_0$, $\mathbf{r}(1) = \mathbf{p}_1$, $\mathbf{r}'(1) = \mathbf{v}_1$ and $\mathbf{r}''(1) = \mathbf{a}_1$. This Hermite interpolant is given by

$$\mathbf{r}_c(u) = (9u^5 - 20u^4 + 10u^3 + 1, -9u^5 + 25u^4 - 20u^3 + 5u) \quad (1)$$

and shown in Figure 1. However this interpolant assuring C^2 continuity has a loop and is not suitable for rounding in CNC applications.

Relaxing C^2 to G^2 continuity gives more freedom and allows for optimizing a cost function. Using the approach explained in Section 4 we obtain the following optimal degree 5 polynomial yielding G^2 continuity

$$\mathbf{r}_g(u) = \left(-\frac{1}{6}u^5 + \frac{5}{6}u^4 - \frac{5}{3}u^3 + 1, \frac{1}{6}u^5 - \frac{5}{3}u^2 + \frac{5}{2}u\right). \quad (2)$$

Note that $\mathbf{r}'_g(0) = (0, 5/2)$ which is collinear to $\mathbf{v}_0$. Furthermore, $\mathbf{r}''_g(0) = (0, -10/3)$, hence curvature κ_0 at $\mathbf{p}_0$ vanishes, assuring geometric G^2 continuity.

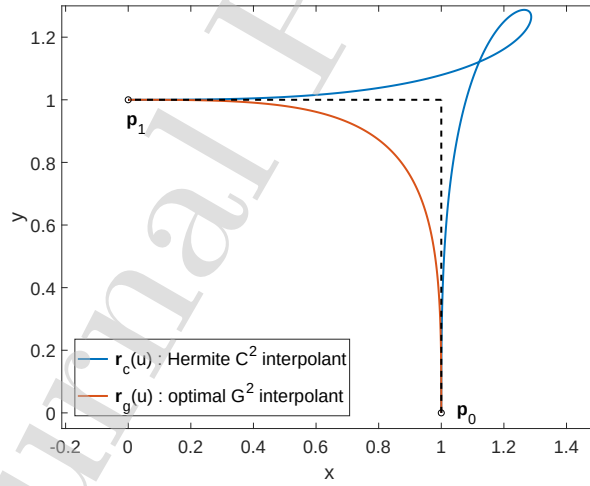


Figure 1: Comparison between C^2 Hermite and optimal G^2 interpolant

3. Problem setup

Consider a curve in $\mathbb{R}^3$ with pieces $\mathcal{C}_0, \mathcal{C}_1$ having a G^2 discontinuous junction point $\mathbf{p}_j$. In order to smooth the transition, we will cut out a part of $\mathcal{C}_0$ and $\mathcal{C}_1$,

and replace the missing parts by a G^2 continuous transition curve $\mathcal{C}$. We will call the new junctions “lift-off points” $\mathbf{p}_0, \mathbf{p}_1$ according to Figure 2.

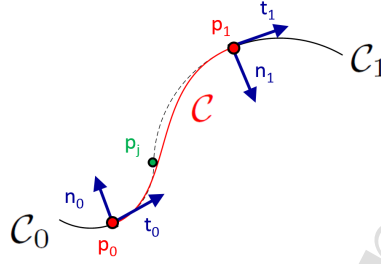


Figure 2: Problem setup

Consider the orthonormal Frenet frames $(\mathbf{t}_0, \mathbf{n}_0, \mathbf{b}_0)$, $(\mathbf{t}_1, \mathbf{n}_1, \mathbf{b}_1)$ and curvatures κ_0, κ_1 defined at both lift-off points as

$$\mathbf{t} = \frac{\mathbf{r}'}{\|\mathbf{r}'\|}, \quad \mathbf{b} = \frac{\mathbf{r}' \times \mathbf{r}''}{\|\mathbf{r}' \times \mathbf{r}''\|}, \quad \mathbf{n} = \mathbf{b} \times \mathbf{t}, \quad \kappa = \frac{\|\mathbf{r}' \times \mathbf{r}''\|}{\|\mathbf{r}'\|^3}. \quad (3)$$

According to [13], geometric G^2 continuity at the lift-off points is achieved if the following holds

$$\mathbf{r}(0) = \mathbf{p}_0 \quad (4)$$

$$\mathbf{r}(1) = \mathbf{p}_1 \quad (5)$$

$$\mathbf{r}'(0) = \alpha_0 \mathbf{t}_0 \quad (6)$$

$$\mathbf{r}'(1) = \alpha_1 \mathbf{t}_1 \quad (7)$$

$$\alpha_0 > 0 \quad (8)$$

$$\alpha_1 > 0 \quad (9)$$

$$\mathbf{r}''(0) = \beta_0 \mathbf{t}_0 + \gamma_0 \mathbf{n}_0 \quad (10)$$

$$\gamma_0 = \kappa_0 \alpha_0^2 \quad (11)$$

$$\mathbf{r}''(1) = \beta_1 \mathbf{t}_1 + \gamma_1 \mathbf{n}_1. \quad (12)$$

$$\gamma_1 = \kappa_1 \alpha_1^2 \quad (13)$$

The feasibility conditions (8), (9) ensure a cusp-free transition curve. Infinitely many curves achieve G^2 continuity. In order to select the “best” one, we introduce the following objective function to be minimized

$$J = \int_0^1 \|\mathbf{r}'''(u)\|^2 du. \quad (14)$$

This choice of objective function J is motivated as follows. It ensures that $\mathbf{r}''(u)$ is continuous, it leads to a tractable computational problem as will be shown below, and it translates somehow the fairness of the interpolant. As an example, the value of the cost function J is 3840 for the C^2 interpolant

defined in Equation (1), and only 40 for the much better G^2 interpolant in Equation (2). Another choice for the objective function would have been the integral of squared derivative of curvature, but this choice would have led to a much more difficult mathematical problem.

Note that the minimization problem (14) under G^2 continuity constraints (4), ..., (13) is a variational problem with an unknown function $\mathbf{r}(u)$ and scalar unknowns $\alpha_0, \alpha_1, \beta_0, \beta_1, \gamma_0, \gamma_1$. The quadratic constraints (11), (13) render the problem non-convex.

4. Problem solution

The optimization problem described above can be solved numerically using software packages as [14], [15]. However, a good initial guess is needed, and calculation time is quite high. This motivates the following more efficient analytical solution.

Theorem 1. Any solution of the variational problem (14) under G^2 Hermite continuity constraints (4), ..., (13) is given by a parametric curve defined by quintic polynomials.

Proof. Let $\mathbf{r}(u)$ be a solution of our problem and let us consider an additive perturbation $\boldsymbol{\eta}(u)$ as used in standard variational calculus. We call $\boldsymbol{\eta}(u)$ admissible if $\mathbf{r}(u) + \boldsymbol{\eta}(u)$ respects de G^2 continuity constraints. Since $\mathbf{r}(u)$ is optimal, we have

$$\int_0^1 \|\mathbf{r}'''(u)\|^2 du \leq \int_0^1 \|\mathbf{r}'''(u) + \boldsymbol{\eta}'''(u)\|^2 du$$

for all admissible $\boldsymbol{\eta}(u)$. Now, observe that for the special case of all admissible $\boldsymbol{\eta}(u)$ whose derivatives vanish on the boundaries, $\mathbf{r}(u) + \lambda \boldsymbol{\eta}(u)$ is admissible for all $\lambda \in \mathbb{R}$. Define the function

$$\begin{aligned} f(\lambda) &:= \int_0^1 \|\mathbf{r}'''(u) + \lambda \boldsymbol{\eta}'''(u)\|^2 du \\ &= \int_0^1 \|\mathbf{r}'''(u)\|^2 du + 2\lambda \int_0^1 \mathbf{r}'''(u) \cdot \boldsymbol{\eta}'''(u) du + \lambda^2 \int_0^1 \|\boldsymbol{\eta}'''(u)\|^2 du \end{aligned}$$

where $\cdot$ denotes the scalar product. This function satisfies $f'(0) = 0$, since $\mathbf{r}(u)$ is supposed to be optimal. Hence,

$$\int_0^1 \mathbf{r}'''(u) \cdot \boldsymbol{\eta}'''(u) du = 0 \quad (15)$$

for all admissible $\boldsymbol{\eta}(u)$ whose derivatives vanish on the boundaries. Applying the integration by parts rule three times, Equation (15) becomes

$$\int_0^1 \mathbf{r}^{(6)}(u) \cdot \boldsymbol{\eta}(u) du = 0, \quad (16)$$

where we use the superscript $\cdot^{(n)}$ for the n^{th} parametric derivative. Therefore, $\mathbf{r}^{(6)}(u) = \mathbf{0}$, and the components of $\mathbf{r}(u)$ are polynomials of degree (at most) 5. Note that quintic B-splines with piecewise *different* polynomials of degree 5 can be excluded as solution. The discontinuities in the derivatives would make appear "Dirac peaks" (distributions). A detailed mathematical complement to the proof of Theorem 1 is given in Appendix A. $\square$

Theorem 2. The determination of the polynomial coefficients of the solution curve of (14) under G^2 Hermite continuity constraints (4), ..., (13) leads to finding the positive roots of a polynomial system with two unknowns α_0, α_1 . The resultant of this system is given by a polynomial of degree ≤ 9 , yielding at most 9 stationary points.

Proof. Let $\mathbf{r}(u)$ be a feasible solution of our problem, and let $\boldsymbol{\eta}(u)$ be the unique polynomial of degree 5 such that

$$\begin{aligned}\boldsymbol{\eta}(0) &= \mathbf{0}, \quad \boldsymbol{\eta}(1) = \mathbf{0}, \quad \boldsymbol{\eta}'(0) = \delta_0 \mathbf{t}_0, \quad \boldsymbol{\eta}'(1) = \delta_1 \mathbf{t}_1, \\ \boldsymbol{\eta}''(0) &= \epsilon_0 \mathbf{t}_0 + \zeta_0 \mathbf{n}_0, \quad \boldsymbol{\eta}''(1) = \epsilon_1 \mathbf{t}_1 + \zeta_1 \mathbf{n}_1\end{aligned}$$

where

$$\zeta_0 = \kappa_0(2\alpha_0\delta_0 + \delta_0^2), \quad \zeta_1 = \kappa_1(2\alpha_1\delta_1 + \delta_1^2).$$

It is not difficult to check that $\mathbf{r}(u) + \boldsymbol{\eta}(u)$ is admissible for $|\delta_0|$ and $|\delta_1|$ sufficiently small. Note that this choice of perturbation $\boldsymbol{\eta}(u)$ is more general than the special case utilized for the proof of Theorem 1. We now introduce the function

$$f(\delta_0, \delta_1, \epsilon_0, \epsilon_1) := \int_0^1 \|\mathbf{r}'''(u) + \boldsymbol{\eta}'''(u)\|^2 du$$

and using integrations by parts, together with the fact that $\mathbf{r}(u)$ is a polynomial of degree 5, we get

$$\begin{aligned}f(\delta_0, \delta_1, \epsilon_0, \epsilon_1) &= \\ &\int_0^1 \|\mathbf{r}'''(u)\|^2 du + 2 \left(\mathbf{r}'''(u) \cdot \boldsymbol{\eta}''(u) \Big|_0^1 - \mathbf{r}''''(u) \cdot \boldsymbol{\eta}'(u) \Big|_0^1 \right) + \int_0^1 \|\boldsymbol{\eta}'''(u)\|^2 du.\end{aligned}$$

Since $\mathbf{r}(u)$ is optimal, $\mathbf{grad} f(0, 0, 0, 0) = \mathbf{0}$, and this leads to the following equations

$$\mathbf{r}^{(3)}(0) \cdot \mathbf{t}_0 = 0 \tag{17}$$

$$\mathbf{r}^{(3)}(1) \cdot \mathbf{t}_1 = 0 \tag{18}$$

$$2\kappa_0\alpha_0\mathbf{r}^{(3)}(0) \cdot \mathbf{n}_0 - \mathbf{r}^{(4)}(0) \cdot \mathbf{t}_0 = 0 \tag{19}$$

$$2\kappa_1\alpha_1\mathbf{r}^{(3)}(1) \cdot \mathbf{n}_1 - \mathbf{r}^{(4)}(1) \cdot \mathbf{t}_1 = 0. \tag{20}$$

Knowing from Theorem 1 that the solution $\mathbf{r}(u)$ is a degree 5 polynomial, we impose G^2 continuity conditions Equations (4), ..., (7), (10), ..., (13) using the following Hermite interpolation

$$\begin{aligned} \mathbf{r}(u) = & \mathbf{p}_0 h_{00}(u) + \alpha_0 \mathbf{t}_0 h_{10}(u) + (\beta_0 \mathbf{t}_0 + \kappa_0 \alpha_0^2 \mathbf{n}_0) h_{20}(u) + \\ & \mathbf{p}_1 h_{01}(u) + \alpha_1 \mathbf{t}_1 h_{11}(u) + (\beta_1 \mathbf{t}_1 + \kappa_1 \alpha_1^2 \mathbf{n}_1) h_{21}(u) \end{aligned} \quad (21)$$

with the quintic Hermite basis functions

$$\begin{aligned} h_{00}(u) &= (6u^2 + 3u + 1)(1 - u)^3 \\ h_{10}(u) &= (3u^2 + u)(1 - u)^3 \\ h_{20}(u) &= \frac{1}{2} u^2 (1 - u)^3 \\ h_{01}(u) &= u^3 (6u^2 - 15u + 10) \\ h_{11}(u) &= u^3 (-3u^2 + 7u - 4) \\ h_{21}(u) &= \frac{1}{2} u^3 (u - 1)^2. \end{aligned} \quad (22)$$

At this stage we have four scalar unknowns $\alpha_0, \beta_0, \alpha_1, \beta_1$ and four equations, the optimality Equations (17), ..., (20). By elimination of β_0 and β_1 , we can reduce to a polynomial system with two unknowns α_0, α_1 having the following structure

$$\begin{aligned} (a_1 \alpha_1 + a_0) \alpha_0^2 + (b_1 \alpha_1 + b_0) \alpha_0 + \\ c_3 \alpha_1^3 + c_2 \alpha_1^2 + c_1 \alpha_1 + c_0 &= 0 \\ (d_1 \alpha_0 + d_0) \alpha_1^2 + (e_1 \alpha_0 + e_0) \alpha_1 + \\ f_3 \alpha_0^3 + f_2 \alpha_0^2 + f_1 \alpha_0 + f_0 &= 0, \end{aligned} \quad (23)$$

where the 16 coefficients $a_0, a_1, \dots, f_0, f_1, f_2, f_3$ depend on the geometrical data $\mathbf{p}_0, \mathbf{t}_0, \mathbf{n}_0, \kappa_0, \mathbf{p}_1, \mathbf{t}_1, \mathbf{n}_1, \kappa_1$. The resultant of both polynomials in Equations (23) is defined as the determinant of their **Sylvester matrix**. In the general case it corresponds here to a monovariate polynomial of degree at most 9, either in α_0 or in α_1 . The global optimum is found by taking the minimum of the objective function over at most nine stationary points. $\square$

The expressions for the 16 coefficients in Equations (23) and the coefficients of the resultant polynomial are quite cumbersome. However, it is easy to derive the equations using any computer algebra system. The Matlab sources of all symbolic and numeric calculations of the algorithm are made available under [16]. The symbolic calculations are carried out only once at the development phase of the algorithm. At run time, all calculations are purely numerical.

5. Rounding algorithm

1. Choose the curve coordinates u_0, u_1 corresponding to the lift-off points $\mathbf{p}_0, \mathbf{p}_1$ on either side of the original non-smooth junction point $\mathbf{p}_j$. Section 7 explains this choice more in detail.

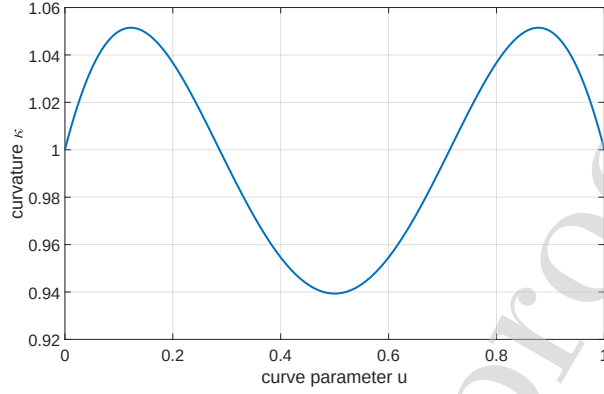
2. Calculate the local Frenet frames $(\mathbf{t}_0, \mathbf{n}_0, \mathbf{b}_0)$, $(\mathbf{t}_1, \mathbf{n}_1, \mathbf{b}_1)$ and curvatures κ_0, κ_1 at both lift-off points using the parametric expressions of the abutted curves.
3. Calculate numerically the 16 coefficients $a_0, a_1, \dots, f_0, f_1, f_2, f_3$ of the polynomial system in Equations (23).
4. Treat separately the degenerate case $\kappa_0 = \kappa_1 = 0$, where the polynomial system degenerates to a linear system. Solve it, check if $\alpha_0 > 0$, $\alpha_1 > 0$ and go to 10). There is no G^2 continuous transition if $\alpha_0 \leq 0$ or $\alpha_1 \leq 0$. In this case the lift-off points have to be modified, see Section 7.
5. Calculate numerically the coefficients of the resultant polynomial w.r.t. either α_0 or α_1 .
6. Calculate all roots of this polynomial (e.g. using standard algorithms based on computing the eigenvalues of the companion matrix) . Discard complex conjugate and negative real roots.
7. For the remaining values of α_0 (resp. α_1), calculate α_1 (resp. α_0). This amounts to solving a linear equation. Discard negative values.
8. For the remaining positive candidate couples $(\alpha_{0,k}, \alpha_{1,k})$, calculate the solution curve $\mathbf{r}(u)$ by backsubstitution and calculate the cost function for each couple. No numerical integration is needed since the integrand is a polynomial. These candidates correspond to stationary points (either local minima, local maxima, or saddle points).
9. Select the couple $(\alpha_{0,k}^*, \alpha_{1,k}^*)$ yielding the minimum cost function.
10. Backsubstitute in order to get the solution curve $\mathbf{r}(u)$.

A simple numerical example of the proposed rounding algorithm is given in Section 2, Equation (2), depicted in Figure 1.

The proposed rounding algorithm either finds the global optimum, or detects an unfeasibility situation if the lift-off points are not well chosen, refer to Section 7. The interesting feature is the assurance of finding the global solution without need of gridding and without using an iterative procedure needing an initial guess. This is important for industrial CNC applications where huge amounts of adjacent curve pieces have to be rounded in a fast and reliable manner. Regarding calculation duration, one rounding programmed in Matlab needs 0.2 milliseconds running on an Intel Core™ i7-3520M @ 2.9 GHz.

6. Circular interpolation example

As a numerical benchmark problem we propose the case of circular interpolation proposed by [10, 11]. Here $\mathbf{p}_0 = (\cos \frac{\alpha}{2}, -\sin \frac{\alpha}{2})$, $\mathbf{t}_0 = (\sin \frac{\alpha}{2}, \cos \frac{\alpha}{2})$, $\mathbf{p}_1 = (\cos \frac{\alpha}{2}, \sin \frac{\alpha}{2})$, $\mathbf{t}_1 = (-\sin \frac{\alpha}{2}, \cos \frac{\alpha}{2})$, and $\kappa_0 = \kappa_1 = 1$. For $\alpha = \frac{\pi}{2}$, the numerical solution of the polynomial system yields $\alpha_0 = \alpha_1 = 1.6633$. Figure 3 shows the curvature of the resulting solution curve.

Figure 3: Curvature as a function of curve parameter u

Clearly, the solution curve is close to a circular arc of radius 1.

As a second example, consider the spiral $\mathbf{r}(u) = (u \cos(u), u \sin(u), u)$ in $\mathbb{R}^3$. We are searching a curve completion between $u_0 = \pi/2$ and $u_1 = 3\pi/2$. The geometric input data are

$$\begin{aligned} \mathbf{p}_0 &= (0, \pi/2, \pi/2), \mathbf{p}_1 = (0, -3\pi/2, 3\pi/2), \\ \mathbf{t}_0 &\approx (-0.7432, 0.4731, 0.4731), \mathbf{t}_1 \approx (0.9578, -0.2033, 0.2033), \\ \mathbf{n}_0 &\approx (-0.5952, -0.7904, -0.1446), \mathbf{n}_1 \approx (0.2153, 0.9758, -0.0387), \\ \kappa_0 &\approx 0.5444, \text{ and } \kappa_1 \approx 0.2077. \end{aligned}$$

The following table shows the 9 roots of the resultant in α_0 , as well as the values of α_1 corresponding to positive roots of α_0 .

α_0	α_1
-6.9784	-
-6.3743	-
-6.0395	-
7.6173	13.2244
6.2695	-5.6170
5.2548	-8.8961
-1.2150	-
1.6221	-10.8066
0.4322	-1.5801

Table 1: Table with relevant roots of the polynomial system.

The only feasible solution in Table 1 is $\alpha_0 = 7.6173$ and $\alpha_1 = 13.2244$. Back-substitution gives the degree 5 polynomial transition shown in red in Figure 4.

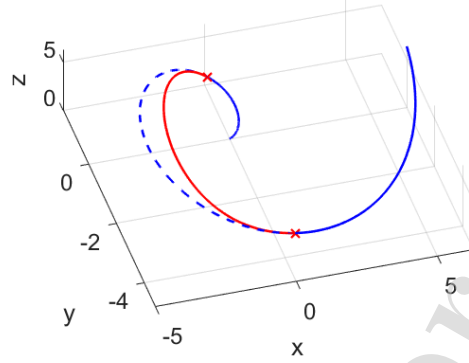


Figure 4: Curve completion for spiral

7. Choice of lift-off points

Figure 5 shows a situation where the polynomial system of Equations (23) degenerates to a singular linear system, leading to infinitely many solutions. This situation normally does not occur since the CNC code ensures an intersection of succeeding curve pieces.



Figure 5: Singular case

Some precautions regarding the choice of the lift-off points must be taken. Figure 6 with lift-off points p_{0b} and p_1 shows a situation where no positive solution for α_0, α_1 exists, leading to a cusp. As a remedy, lift-off point p_{0b} must be shifted to some point p_{0g} on the other side of junction point p_j .

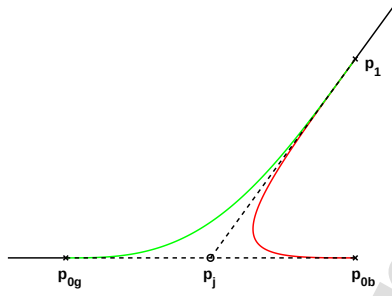


Figure 6: Good and bad choices of lift-off points

Figure 7 shows a helix joining at $\mathbf{p}_j$ an arc of circle lying in a horizontal plane. There is no positive solution for α_0, α_1 for the choice of lift-off points $\mathbf{p}_{0b}$ and $\mathbf{p}_{1b}$ being too far away. The transition curve (red) corresponding to the negative solution of α_0, α_1 is depicted in Figure 7. It goes the wrong way around leading to cusps. The problem disappears with closer lift-off points $\mathbf{p}_{0g}, \mathbf{p}_{1g}$.

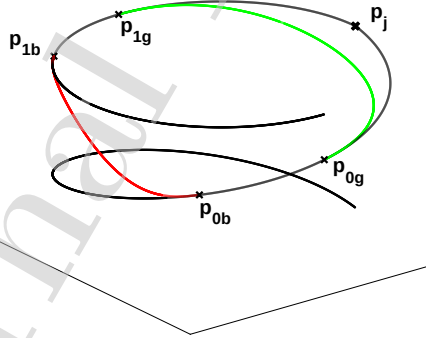


Figure 7: Good and bad choices of lift-off points

Generally, the distance between the lift-off points $\mathbf{p}_0, \mathbf{p}_1$ and the original discontinuous junction point $\mathbf{p}_j$ determines the tolerance between the programmed and the rounded path. This relationship is analyzed in the following section.

8. Tolerance analysis and comparison with clothoids

We consider the case of two line segments intersecting at junction point $\mathbf{p}_j$. Furthermore let us assume that $\|\mathbf{p}_j - \mathbf{p}_0\|_2 = \|\mathbf{p}_j - \mathbf{p}_1\|_2 = d_{\text{cut}}$. The tolerance is given by the distance d_{tol} between $\mathbf{p}_j$ and the rounded solution curve. Since the curvature at lift-off points is zero, the polynomial system (23) degenerates to a linear system.

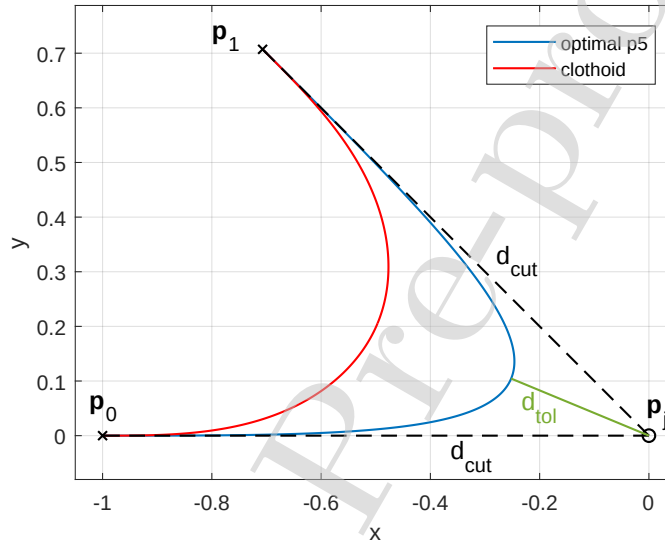


Figure 8: Comparison between optimal transition and clothoid

Therefore, d_{tol} becomes a linear function of d_{cut} . Let α be the opening angle between the line segments. For this easy case, $d_{\text{tol}}(\alpha)$ can be calculated analytically for both cases.

$$d_{\text{tol, poly}}(\alpha) = d_{\text{cut}} \frac{63 \cos\left(\frac{\alpha}{2}\right) + \cos\left(\frac{3\alpha}{2}\right)}{32(\cos(\alpha) + 6)} \quad (24)$$

$$d_{\text{tol, clot}}(\alpha) = d_{\text{cut}} \frac{\int_0^{\sqrt{\pi-\alpha}} \sin\left(\frac{w^2}{2}\right) dw}{\int_0^{\sqrt{\pi-\alpha}} \sin\left(\frac{w^2}{2} + \frac{\alpha}{2}\right) dw} \quad (25)$$

Figure 9 shows the comparison. Clearly, the present method shows less tolerance compared to the clothoid approach. Equation (24) can be used for the proper selection of lift-off points if the tolerance d_{tol} is specified.

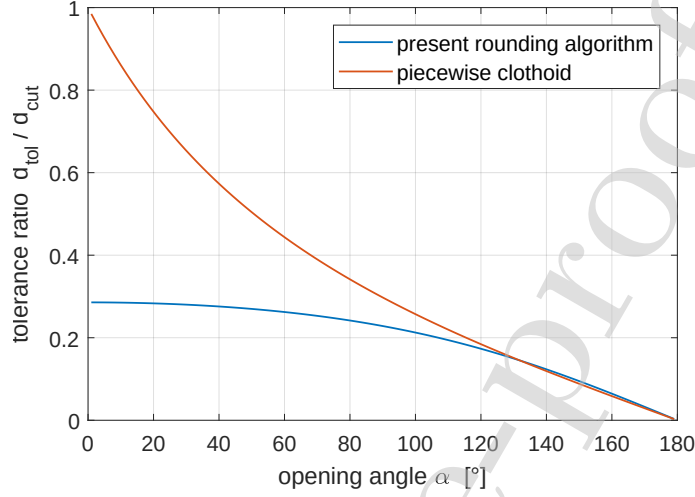


Figure 9: Tolerance analysis

9. Conclusion

The features of the proposed rounding method can be summarized as follows:

- Only one tuning parameter is needed, i.e. the choice of lift-off points. The proximity of lift-off points w.r.t. the discontinuity determines the geometrical tolerance as shown in Section 8.
- The resulting 3D transition curve is a simple quintic polynomial with a natural shape. A polynomial as opposed to a B-Spline or a NURBS is advantageous for CNC controls with limited computation power.
- The coefficient determination for the polynomial transition curve is based on an analytical approach. Necessary optimality conditions are given by the roots of a degree 9 polynomial. Standard well proven algorithms for polynomial root finders can be employed.

Appendix A. Complement to the proof of Theorem 1

The natural functional space to solve problem (4), ... (14) is the set of vector-valued functions whose components belong to the Sobolev space $H^3(0, 1)$, the space of functions $L^2(0, 1)$ whose 1st, 2nd and 3rd derivative (in the sense of distributions) also belong to $L^2(0, 1)$. Note that the embedding $H^3(0, 1) \subset C^2[0, 1]$

is continuous so that conditions (4), $\dots$, (7), (10) and (12) are well-defined. Let $\mathbf{r}$ be optimal and let $\boldsymbol{\eta} \in H^3(0, 1)$ be a special admissible perturbation whose first and second classical derivatives vanish on the boundaries. For $\lambda \in \mathbb{R}$, the function $\mathbf{r} + \lambda \boldsymbol{\eta}$ respects the G^2 constraints and we have

$$\int_0^1 \|\mathbf{r}'''(u)\|^2 du \leq \int_0^1 \|\mathbf{r}'''(u) + \lambda \boldsymbol{\eta}'''(u)\|^2 du$$

and the function

$$\begin{aligned} f(\lambda) &:= \int_0^1 \|\mathbf{r}'''(u) + \lambda \boldsymbol{\eta}'''(u)\|^2 du = \int_0^1 \|\mathbf{r}'''(u)\|^2 du \\ &\quad + 2\lambda \int_0^1 \mathbf{r}'''(u) \cdot \boldsymbol{\eta}'''(u) du + \lambda^2 \int_0^1 \|\boldsymbol{\eta}'''(u)\|^2 du \end{aligned}$$

satisfies $f'(0) = 0$, which implies that

$$\int_0^1 \mathbf{r}'''(u) \cdot \boldsymbol{\eta}'''(u) du = 0.$$

This means precisely that the third derivative $\mathbf{r}'''$, in the sense of distributions, is equal to zero and therefore $\mathbf{r}^{(6)} = \mathbf{0}$. For $0 \leq u_1 < u_2 \leq 1$ we get

$$\mathbf{r}^{(5)}(u_2) - \mathbf{r}^{(5)}(u_1) = \int_{u_1}^{u_2} \mathbf{r}^{(6)}(u) du = \mathbf{0}$$

and hence $\mathbf{r}^{(5)}$ is constant and the components of $\mathbf{r}$ are polynomials of degree (at most) 5.

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10

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subject : highlights of our paper "Optimal G^2 Hermite interpolation for 3D curves"

- The problem of G^2 curve rounding or curve completion in 3D space is important in CAD-CAM and CNC.
- Nonconvex variational problem solved analytically.
- Stationary points given by the positive roots of a degree 9 polynomial.
- No tuning parameters are needed.
- Comparisons with other methods are given.
- The proposed algorithm is well tested in many different situations.

$$\min_{\mathbf{r}(\cdot)} \int_0^1 ||\mathbf{r}'''(u)||^2 du.$$

